

Unit 1: J2EE And JDBC

1 Mark

1. Give full form of JDBC.
2. Give full form of ODBC.
3. Give full form of JNDI.
4. Give full form of API.
5. What does CRUD stand for?
6. Name any two types of JDBC drivers.
7. MySQL Connector/J driver is which type of driver?
8. What is a Web Container?
9. What is PreparedStatement?
10. What is ResultSet?

3 Marks

1. Differentiate between Two-Tier and Three-Tier architecture.
2. Explain ResultSet in JDBC.
3. Explain PreparedStatement with example.
4. Explain CallableStatement with example.
5. Explain Connection, Statement and ResultSet interfaces.
6. Explain Type-3 JDBC driver.
7. Explain Tomcat as a Web Container.

5 Marks

1. Explain JDBC Architecture with diagram.
2. Explain types of JDBC drivers.
3. Explain JDBC-MySQL connectivity stepwise with example.
4. Explain J2EE Architecture with its features.
5. Explain Enterprise Architecture styles.
6. Explain creating a simple JDBC CRUD application.

Unit 2: Servlet

1 Mark

1. What is a Servlet?
2. Which file is known as deployment descriptor?
3. Which servlet life cycle method is called only once?
4. What is the purpose of `web.xml`?
5. What are `doGet()` and `doPost()` methods?
6. What is a Cookie?
7. What is URL rewriting?
8. What is Session?
9. What is RequestDispatcher?
10. Name any two session tracking techniques.

3 Marks

1. Explain Servlet API.
2. Explain RequestDispatcher interface.
3. Explain `include()` and `forward()` methods.

4. Differentiate between `doGet()` and `doPost()`.
5. Explain URL rewriting and hidden form fields.
6. Explain Cookie in session tracking.
7. Explain Session with example.
8. Explain Servlet configuration using deployment descriptor.

5 Marks

1. Explain Servlet Life Cycle with diagram.
2. Explain session tracking approaches in Servlet.
3. Write a servlet code for page counter.
4. Explain handling Servlet request and response.
5. Explain Servlet configuration and deployment.
6. Explain architecture of a Servlet.
7. Explain reading initialization parameters in Servlet.

Unit 3: JSP

1 Mark

1. Give full form of JSP.
2. Which tag allows writing Java code into JSP file?
3. Give syntax of page directive.
4. What is JavaBean?
5. List any two JSP implicit objects.
6. Name JSP scripting elements.
7. What is JSP page scope?
8. What is JSP session scope?
9. What is Expression tag?
10. What is JSP directive?

3 Marks

1. Differentiate between Servlet and JSP.
2. Explain JSP scripting elements.
3. Explain JSP implicit objects.
4. Explain JSP object scopes.
5. Explain JSP directive elements.
6. Explain JSP exception handling.
7. Explain JSP action elements.

5 Marks

1. Explain JSP Life Cycle.
2. Explain JSP Architecture.
3. Explain JSP Action Elements with examples.
4. Write code for insert, update, delete and display in JSP with MySQL.
5. Explain JavaBeans properties and methods.
6. Explain JSP translation, compilation and execution phases.
7. Explain JSP directives: page, include and taglib.

Unit 4: EJB, MVC and Hibernate

1 Mark

1. Give full form of MVC.
2. Give full form of ORM.
3. Give full form of HQL.
4. Give full form of POJO.
5. Give full form of JTA.
6. Name any one type of EJB.
7. What are the three components of MVC?
8. What is Hibernate Session?
9. Which is Hibernate's own query language?
10. What is EJB?

3 Marks

1. Explain EJB in detail.
2. Explain benefits of EJB.
3. Define MVC with example.
4. Explain Hibernate mapping.
5. Explain Hibernate Annotation.
6. Explain Hibernate configuration file.
7. Explain Hibernate Query Language.
8. Explain primary role of Hibernate Session.

5 Marks

1. Explain MVC Architecture with diagram.
2. Explain Hibernate Architecture in detail.
3. Explain Hibernate architecture with SessionFactory, Session and Transaction.
4. Explain Hibernate configuration file and mapping file with example.
5. Explain Hibernate annotations in detail.
6. Explain Object Relational Mapping with Hibernate.
7. Explain types of EJB.

Unit 5: Spring Framework and Spring Boot

1 Mark

1. Give full form of AOP.
2. Give full form of IoC.
3. Give full form of DI.
4. What is Dependency Injection?
5. What is Spring Bean?
6. What is the role of Core Container?
7. State one advantage of Spring Boot.
8. What is Pointcut?
9. Name any two Spring Boot annotations.
10. What is @RestController?

3 Marks

1. Explain Inversion of Control.
2. Explain Dependency Injection.
3. Explain Spring Bean and its purpose.
4. Explain ApplicationContext and BeanFactory.
5. Explain AOP terminology: Advice, Join Point, Pointcut and Aspect.
6. Explain Spring MVC with suitable example.
7. Explain Spring Context.
8. Explain auto-configuration in Spring Boot.

5 Marks

1. Explain Spring Framework Architecture with modules.
2. Explain Spring MVC Architecture and request flow.
3. Explain Spring MVC components: DispatcherServlet, Controller and ViewResolver.
4. Explain Spring Boot Architecture.
5. Explain features and advantages of Spring Boot.
6. Explain Spring Boot JDBC using JdbcTemplate.
7. Explain Spring Boot Data JPA CRUD operations.
8. Explain exception handling using `@ExceptionHandler` and `@ControllerAdvice`.